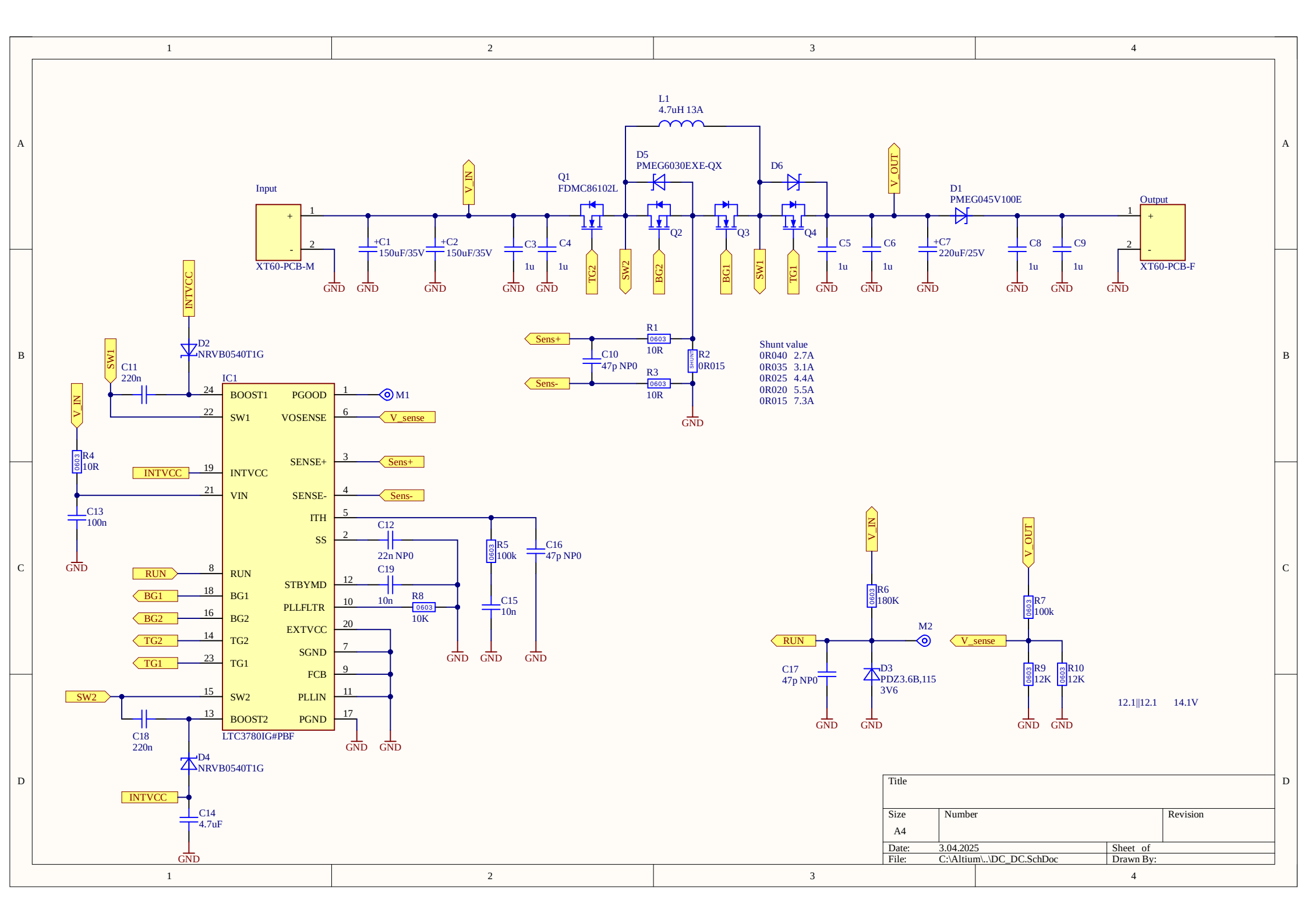
L1
4.7uH 13A
D5
PMEG6030EXE-QX
D6
Q1
FDMC86102L
Q2
Q3
Q4
V_IN
V_OUT
Input
XT60-PCB-M
Output
XT60-PCB-F
D1
PMEG045V100E
+C1
150uF/35V
+C2
150uF/35V
C3
C4
C5
C6
+C7
220uF/25V
C8
C9
1u
GND
TG2
SW2
BG2
BG1
SW1
TG1
Sens+
Sens-
R1
0603
10R
R2
SHUNT
0R015
R3
C10
47p NP0
Shunt value
0R040 2.7A
0R035 3.1A
0R025 4.4A
0R020 5.5A
0R015 7.3A
INTVCC
D2
NRVB0540T1G
C11
220n
IC1
BOOST1
PGOOD
M1
SW1
VOSENSE
V_sense
SENSE+
INTVCC
VIN
SENSE-
ITH
SS
C12
22n NP0
R4
10R
C13
100n
RUN
STBYMD
C19
10n
R8
10K
R5
100k
C16
47p NP0
C15
10n
BG1
BG2
TG2
TG1
PLLFLTR
EXTVCC
SGND
FCB
PLLIN
PGND
SW2
BOOST2
C18
220n
D4
NRVB0540T1G
C14
4.7uF
LTC3780IG#PBF
M2
R6
180K
C17
47p NP0
D3
PDZ3.6B,115
3V6
R7
100k
R9
12K
R10
12K
12.1||12.1
14.1V
Title
Size
A4
Number
Revision
Date: 3.04.2025
Sheet of
File: C:\Altium\..\DC_DC.SchDoc
Drawn By: